

The background of the slide is a composite image. The upper portion shows a tall, red and white drilling rig against a clear blue sky. The lower portion shows an outdoor training facility with various pieces of drilling equipment, including a large crane and several blue and white storage containers, situated in a grassy area with trees in the background.

Introduction to Well Control Equipment and Basic Well Control Technique



Subur Arianto

Introduction

Subur Arianto >>20 years work experiences:



Chevron

DE and DSM for Darajat and Salak, Duri, Minas, and East Kalimantan Offshore.



OTP Geothermal

Assistant Manager Drilling for Sorik Marapi Geothermal Project.



Supreme Energy

Drilling Superintendent and Sr. Drilling Supv for Muara Laboh and Rantau Dedap.



Geo Dipa Energi

Sr. Drilling Supervisor for D2P2 Project, Dieng and Patuha.



Ormat Geothermal

Drilling Site Manager for Toka Tindung.



Sumbawa Timur Mining

Sr. Drilling Supervisor for Hu'u Daha TPBP.



Sarulla Operations Ltd

Sr. Drilling Supervisor for NIL Workover.



Medco Geothermal Sumatera

Sr. Drilling Supervisor for PSPE Bonjol.



Petroleum Training Services Ltd.

Certified Instructor & Assessor: IWCF and IADC, Surface Stack and Subsea Stack.

Why?

- The drilling industry spends millions of dollars on well control related events and incidents.
- Well control event often results in environmental spills that also add to these costs.
- However, the most important thing! Well control problems can lead to a loss of something more valuable than money: “HUMAN LIFE”



Drilling Risk Management

- Identifies **Hazards, Risks** and **Consequences**
- Defines the **Probability** and the **Impact** of each risk
- Avoid the risks or mitigate to reduce the risks to **ALARP** [As Low As Reasonably Practicable] by adding safeguards

Impact →

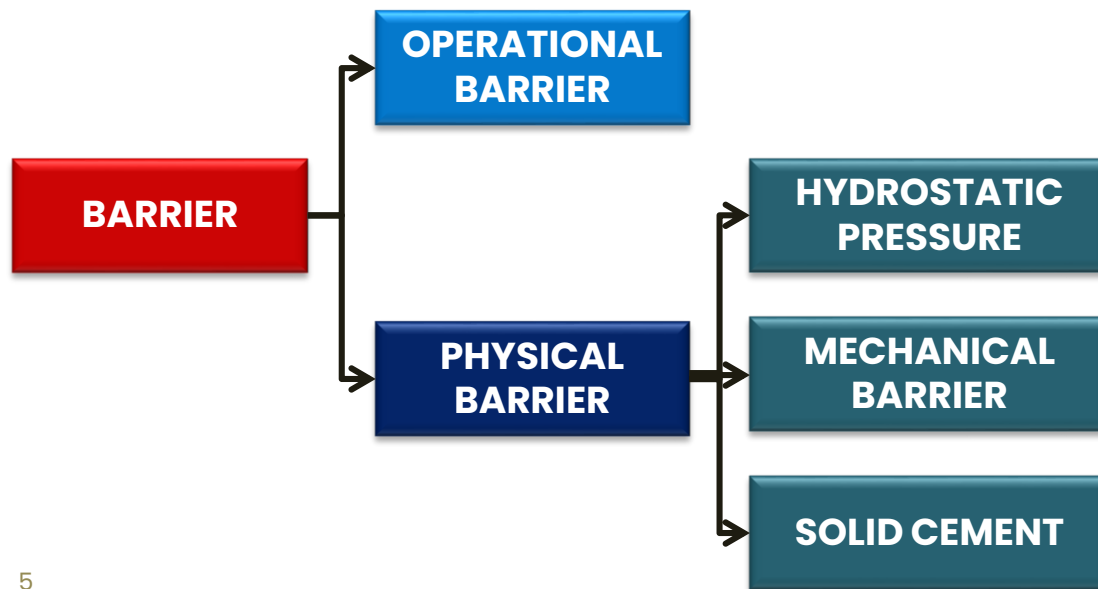
Measures the potential severity of the risk's consequences.

Probability
↑
Measures how likely it is that a risk will occur.

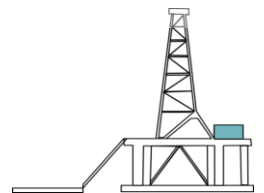
	Insignificant (1)	Minor (2)	Significant (3)	Major (4)	Severe (5)
Almost Certain (5)	5	10	15	20	25
Likely (4)	4	8	12	16	20
Moderate (3)	3	6	9	12	15
Unlikely (2)	2	4	6	8	10
Rarely (1)	1	2	3	4	5

Well Control Safeguards

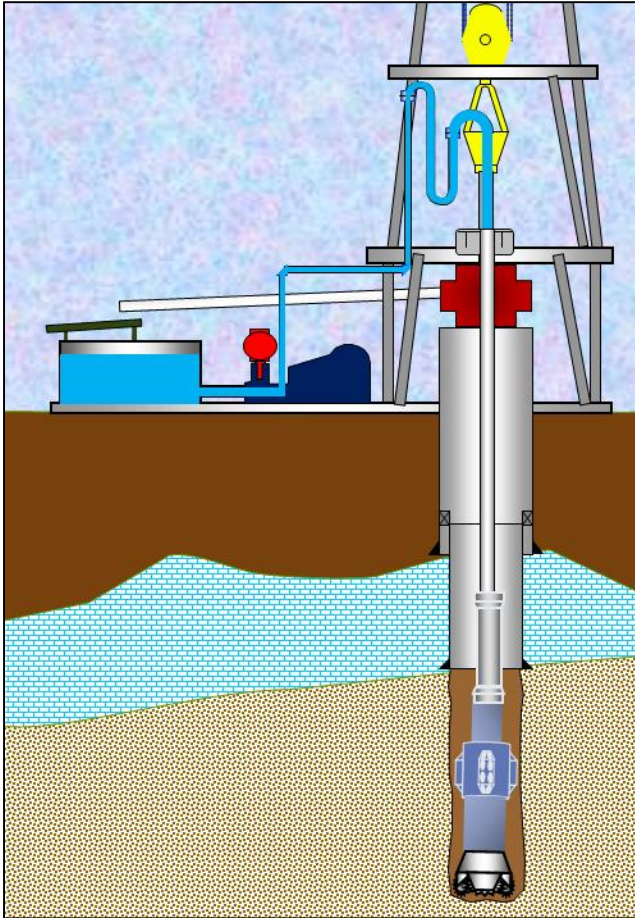
BARRIER: A component and/or practice, if properly deployed, that contributes to the total system reliability in the well by preventing fluid or gas flow to surface.



Well Control Equipment



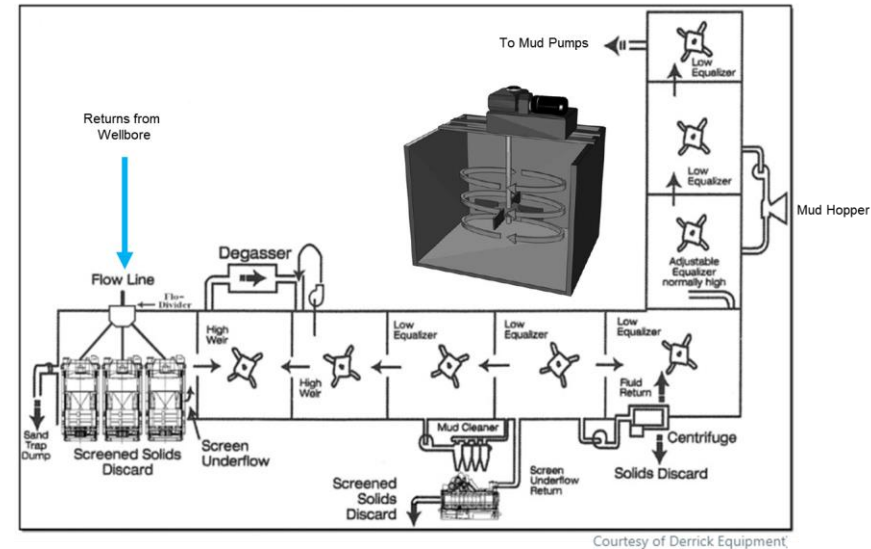
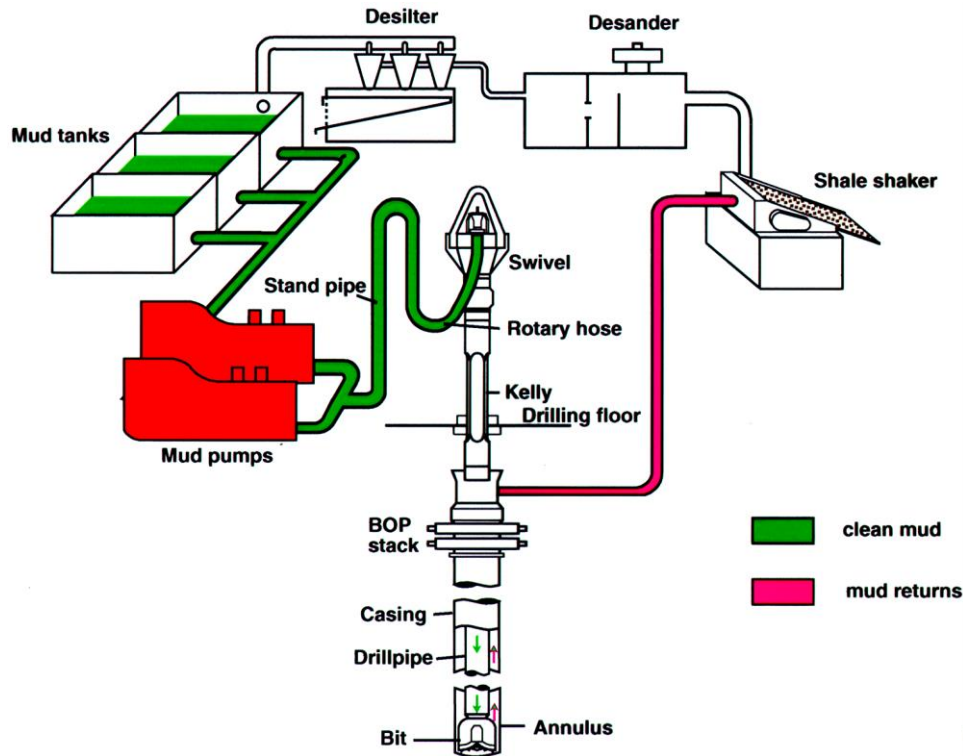
RIG CIRCULATING SYSTEM



The regular closed circulation system circulates drilling fluid

1. From mud tanks,
2. To Mud Pumps,
3. Through the Stand-Pipe, Kelly Hose, Swivel, Kelly (or Top Drive),
4. Down through the drill string, exiting at the bit nozzles,
5. Out to the annulus and up to the surface,
6. Return to the mud tanks via the Flow Line and Shale Shakers.

CIRCULATING SYSTEM



Shaker tank & Mud Tank



Mixing tank

<https://www.gnsolidscontrol.com/mud-tank>



<https://www.nov.com/products/rapid-mud-tank-system>

WELL CONTROL SYSTEM

Major component of Well Control System:

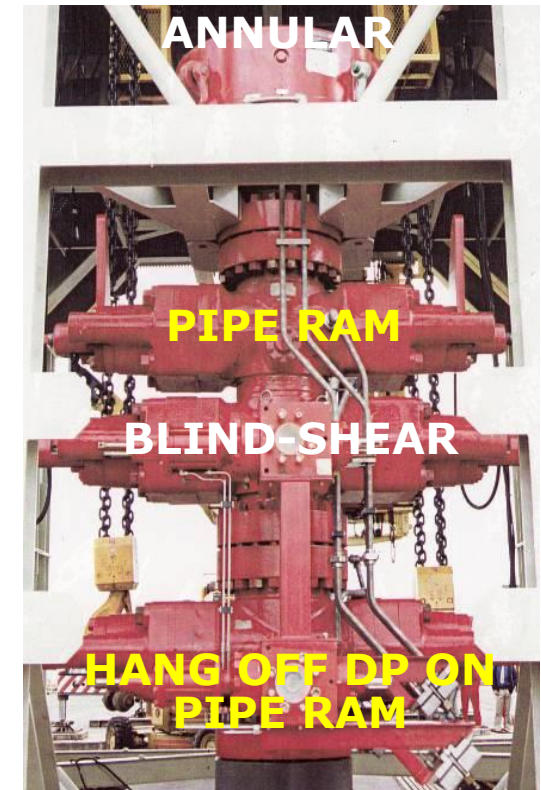
- **BOP Stack:** Annular Preventer, Ram Preventer
- **BOP Control Unit:** Accumulator/Koomey Unit
- **BOP Control Panel:** located closed to driller's panel, remote panel
- **Choke & Kill Lines** including valves
- **Choke Manifold** and valves

BOP STACKS

Any drilling BOP Stack has several vital **functions**:

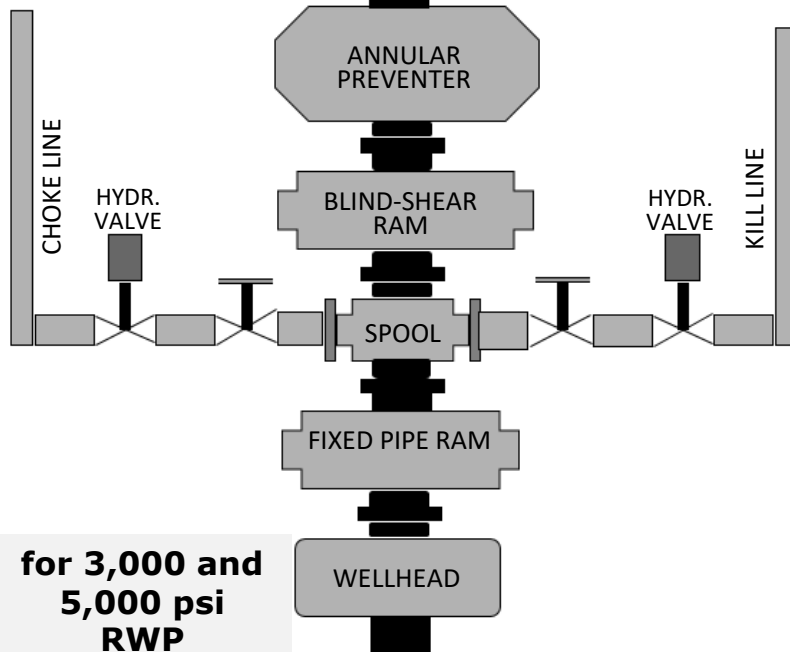
- **Sealing.** A means of shutting in the well around any tubular that happens to be across the BOP.
- **Stripping.** A means of moving pipe [usually drill pipe] in and out with the well under pressure.
- **Blinding.** A means of shutting in the well without any tubular across the BOP. On most rigs, this is combined with a capability to **shear** [drill pipe, casing or both].
- **Hanging off.** A means of hanging of the drill pipe tool joint on one of the pipe rams.

With the side outlets we also have a means of controlled venting [of the influx] and a means of pumping [mud] into a well as required for bullheading and other operations.

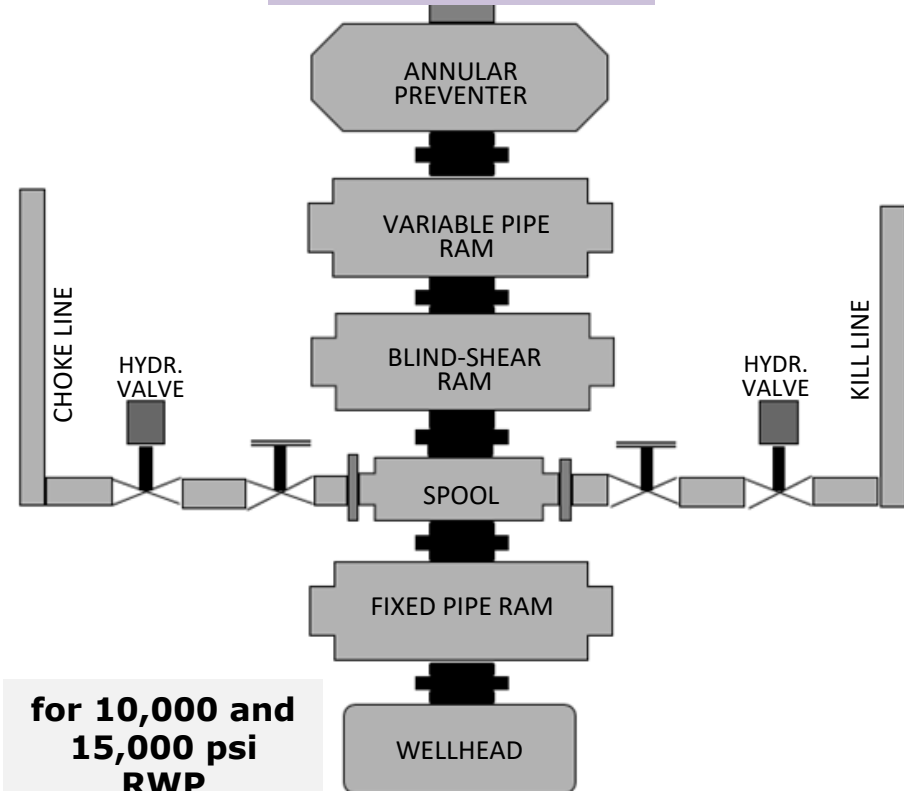


BOP CLASSIFICATION

Class 3 - A1 - R2



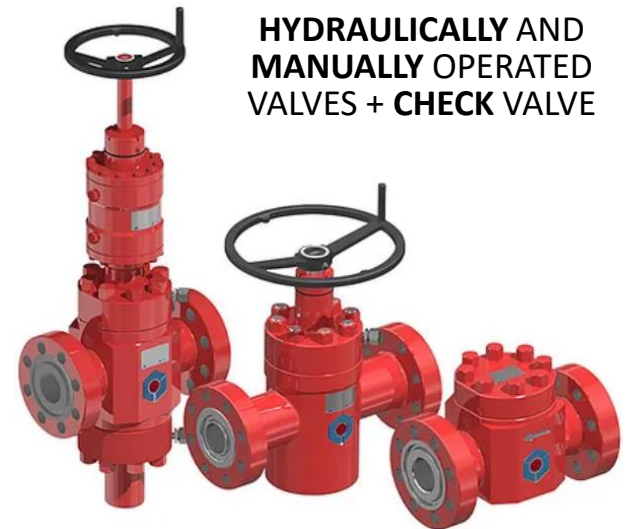
Class 4 - A1 - R3



SIDE OUTLET VALVES

Configurations:

- **Choke lines** must have at least **1 Hydr. Operated Gate Valve**, either on the inside or outside of the choke line.
- **Kill lines** must have at least **1 Hydr. Operated Gate Valve**, either on the inside or outside of the kill line or – alternatively – two manual valves with a check valve.
- A **Check Valve** prevents gas entering the pump area.



HYDRAULICALLY AND
MANUALLY OPERATED
VALVES + CHECK VALVE

DRILLING SPOOL

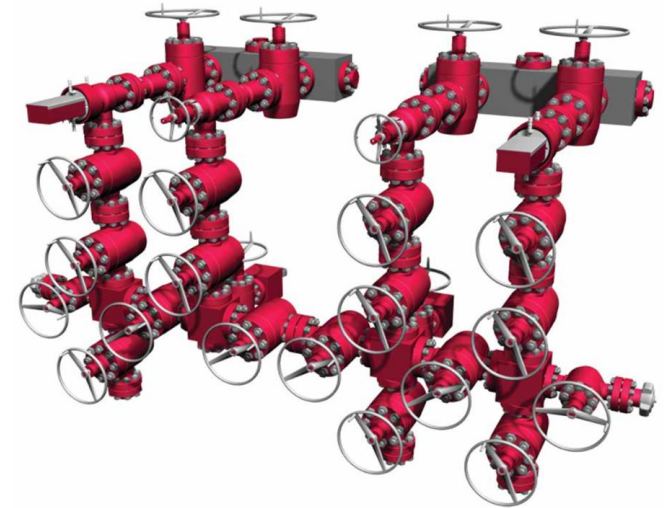
Characteristics:

- The kill- and choke lines can be installed on either the BOPs outlet flanges or on a dedicated drilling spool.
- Using a drilling spool means that any wear will always be on a relatively cheap, **disposable** item, and not on a BOP.
- A spool must have a **RWP** at least equal to that of the Ram BOP.
- The **thru bore** of the spool must be the same as that of the BOP and the max. bore of the uppermost casing/tubing head spool.



CHOKE MANIFOLD

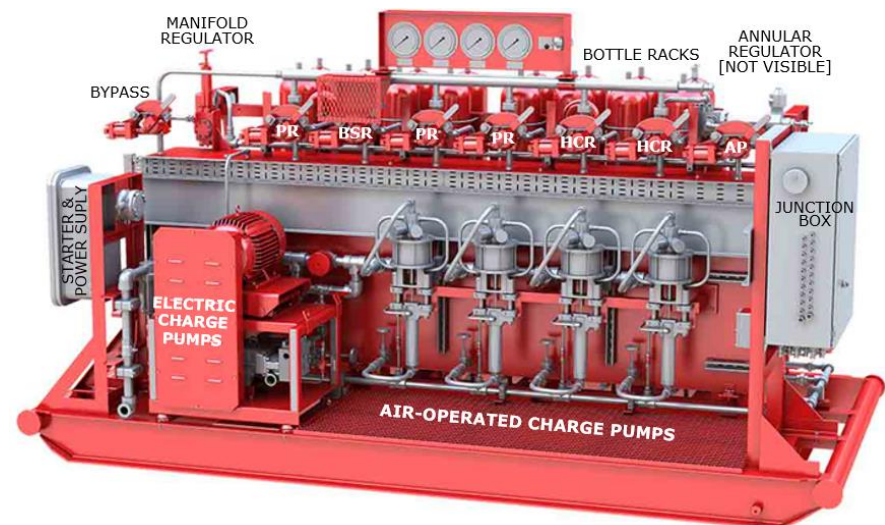
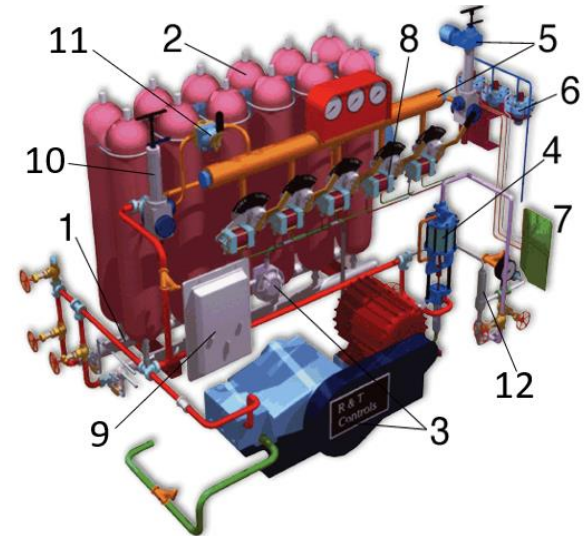
- Up to **4 chokes**, one or two of which can be manual choke, the remaining chokes must be remotely operated.
- Must have a **minimum of 2 chokes**, one of which must be remotely operated. Chokes are considered pressure regulators and not necessarily sealing. There is therefore no requirement to pressure test chokes.
- Must have a **bleed/bypass** line, which enables bleeding off trapped pressure, e.g., after the choke and/or choke line has been plugged off.



BOP HYDRAULIC CONTROL UNIT

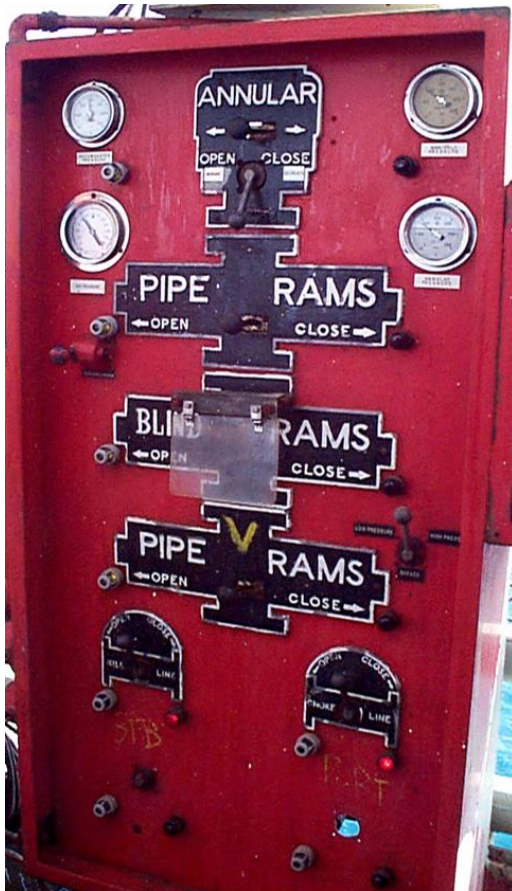
Main Components:

1. Accumulator pressure line.
2. Accumulator bottles.
3. Electrically-operated pump w/ pressure switch.
4. Air-operated pump.
5. Regulator [gear driven] for Annular Preventer.
6. Pressure transmitters.
7. Fixed plate for junction cables.
8. Actuator for manifold **3-position/4-way valve**.
9. Automatic motor starter.
10. Manifold regulator.
11. Bypass [override] valve.
12. Hydro-pneumatic pressure switch.

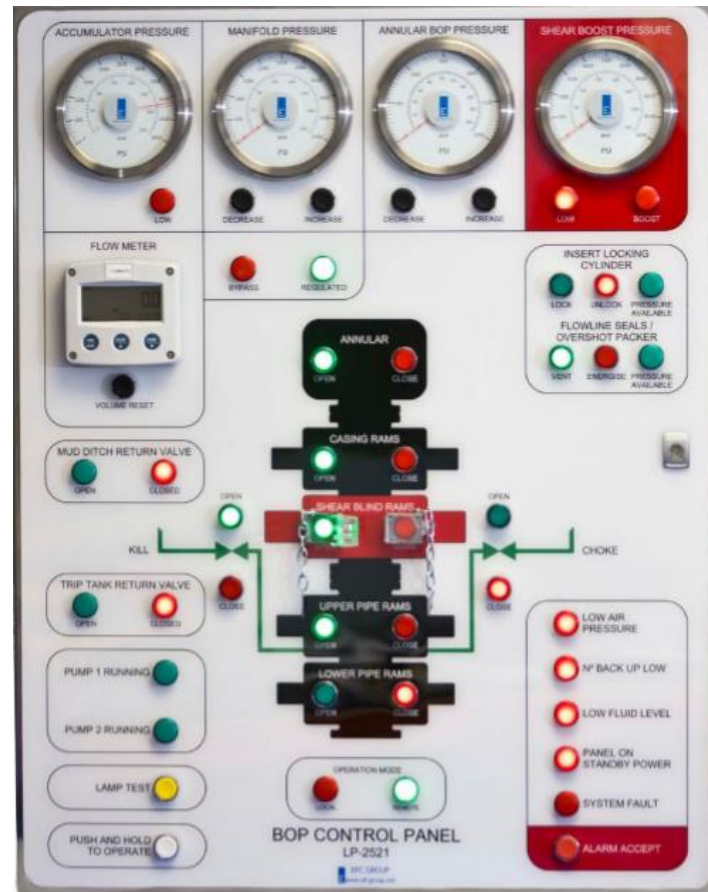


BOP CONTROL PANEL

SURFACE **PNEUMATIC** BOP CONTROL PANEL



SURFACE **ELECTRIC** BOP CONTROL PANEL

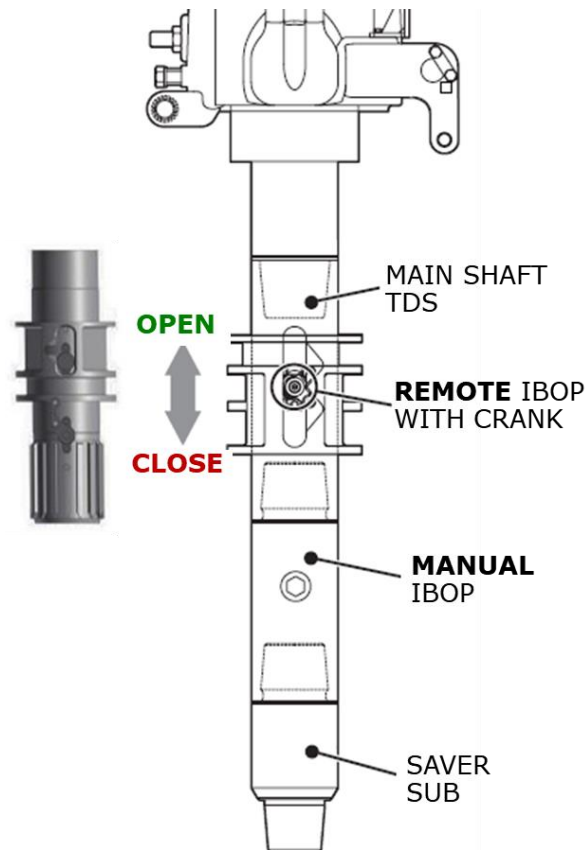


Others

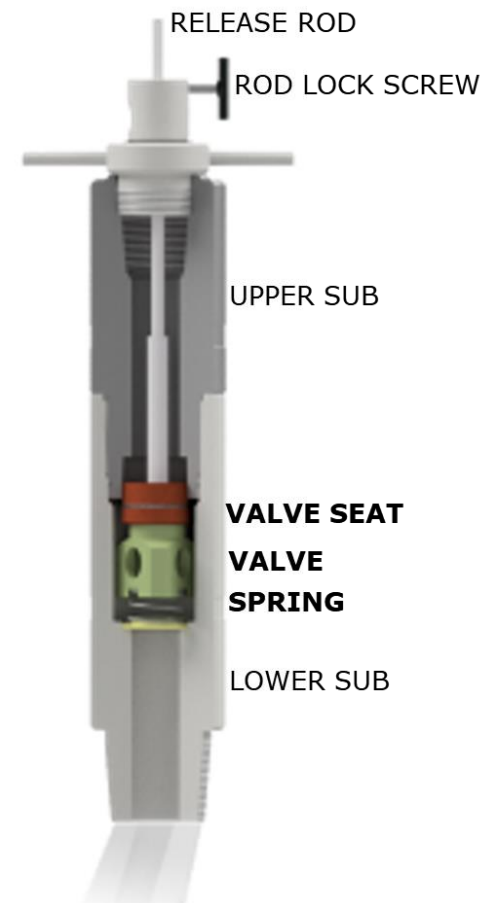
Full Opening Safety Valve (FOSV)



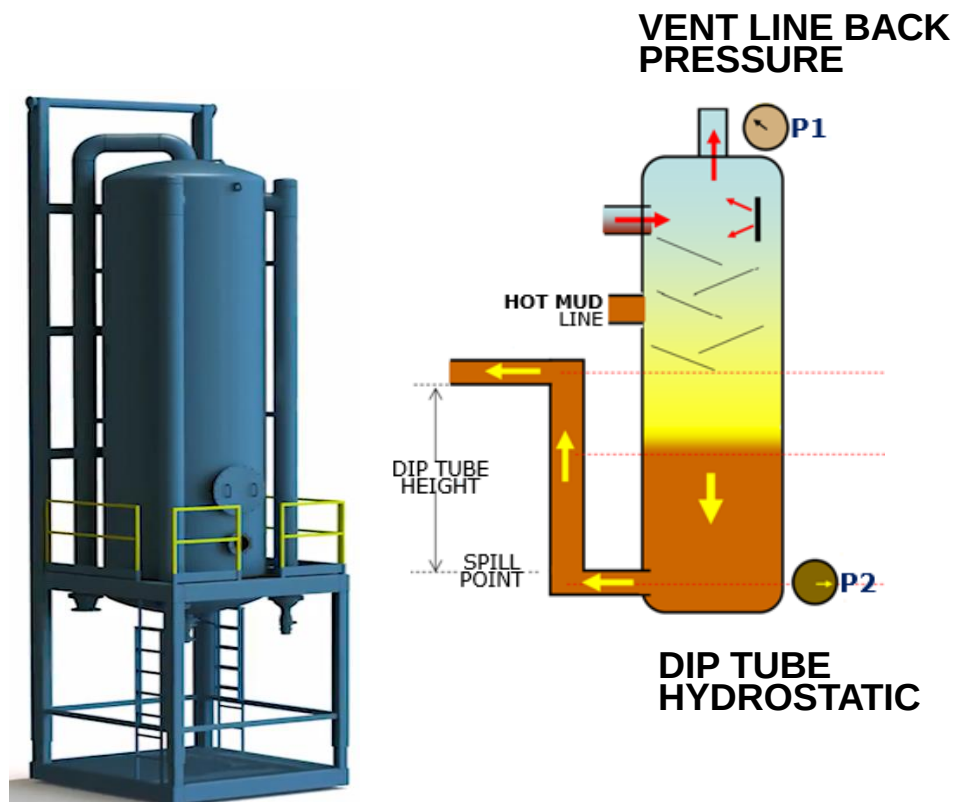
TDS Safety Valve (Top Drive IBOP)



Internal BOP (IBOP)



MUD-GAS SEPARATOR



- Pressure Build Up (back pressure)

- 1- Vent Line ID must be wide
- 2- Vent Line Height Must be short

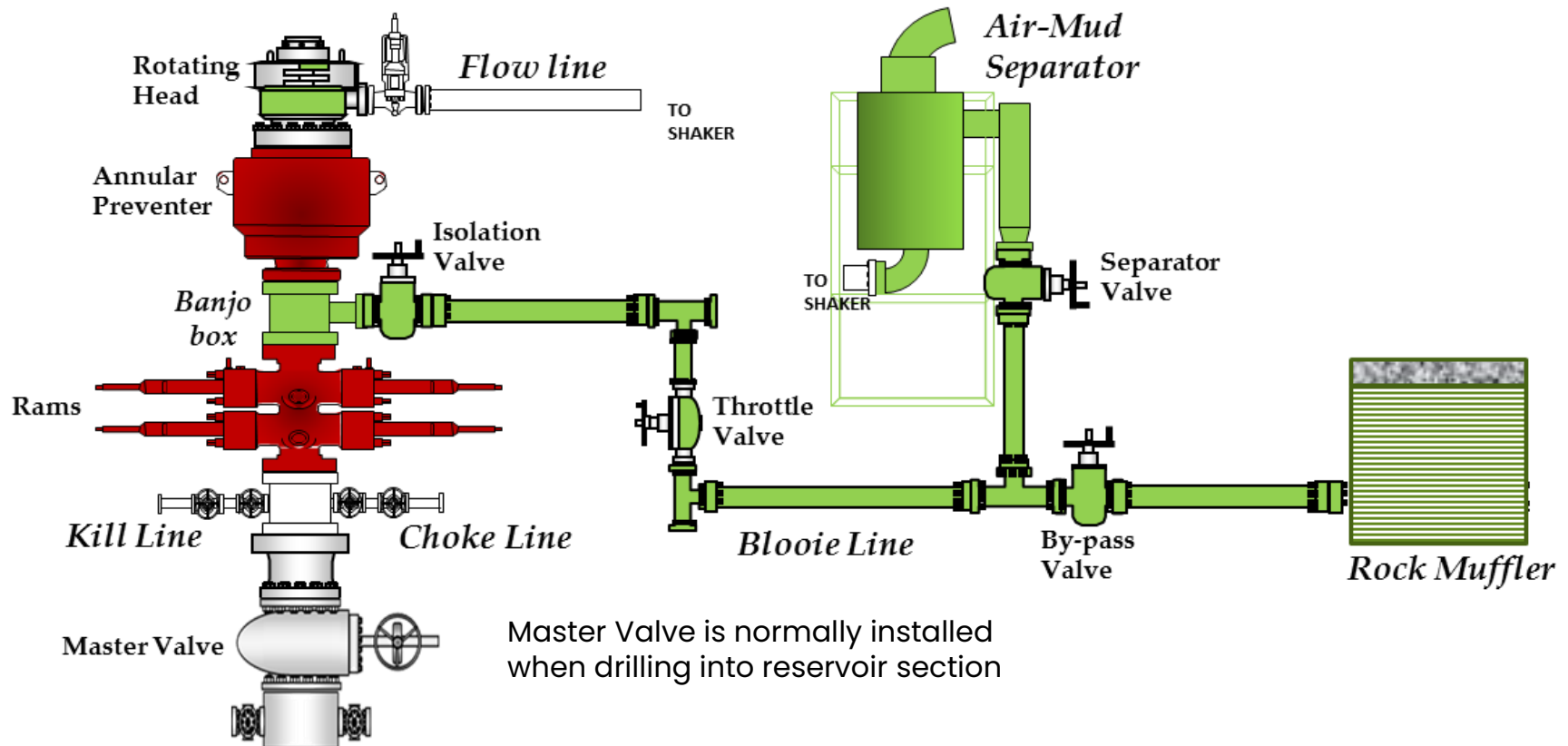
Maximum operating Pressure, normal p. capacity

- 1- U or Dip Tube Height
- 2- Mud Seal weight

Geothermal Well Control

Typical BOP & Air Drilling Set-up

Aerated drilling services can be used when drilling the reservoir section, and may start in the upper hole sections.

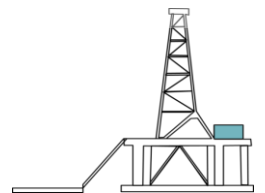


Well Control Basic Concepts





Understanding PRESSURES



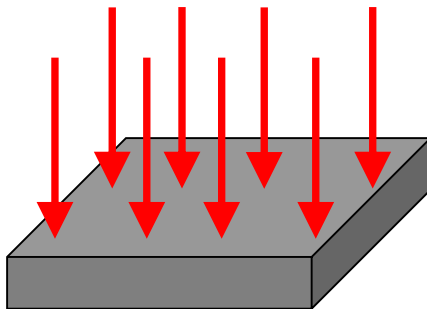
- **What is Pressure?**
- **What is Hydrostatic Pressure?**
- **Why is Hydrostatic Pressure important?**

Pressure

- Pressure = Force exerted per unit Area

$$P = \frac{F}{A}$$

Force in pounds



1 unit area (square inch)

API Units:

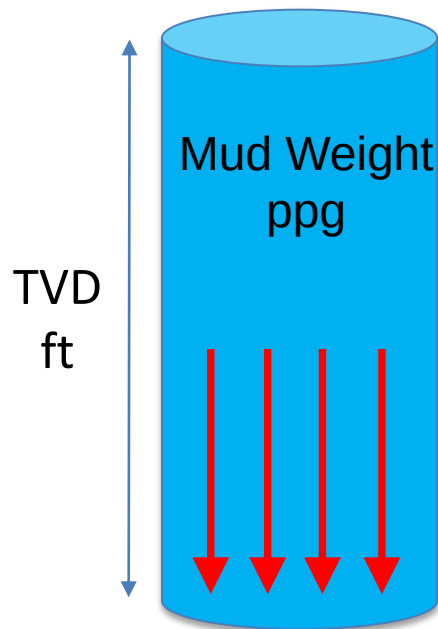
Pounds
Square Inch

(Pounds per Square Inch or 'psi')

Pressure

- **Hydrostatic Pressure**

= pressure caused by the weight of a column of fluid.



- Hydrostatic Pressure (psi) depends on:
 - the weight or density of the fluid (ppg);
 - the vertical height /depth (ft)

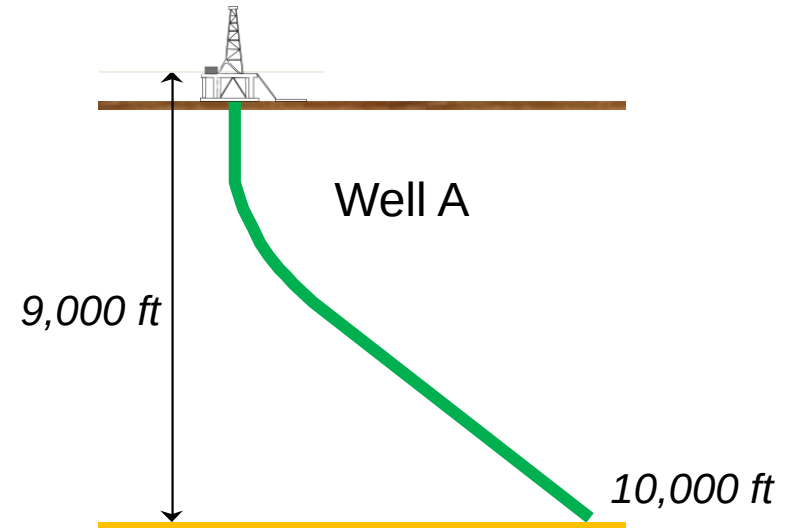
$$\text{Hydrostatic Pressure, psi} = 0.052 \times \text{ppg} \times \text{ft}$$

- The Hydrostatic Pressure per foot depth of a fluid is called Pressure Gradient

Hydrostatic Pressure

- Quiz

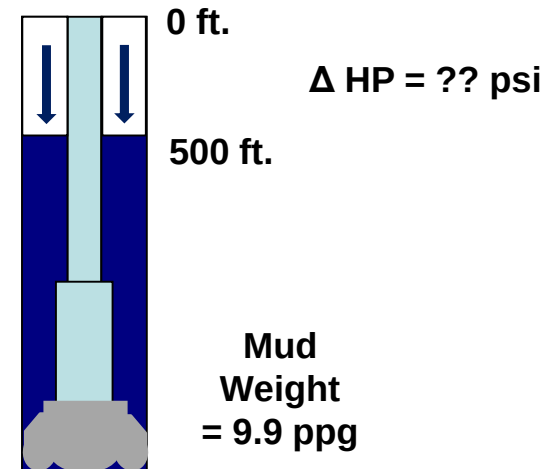
What is the hydrostatic pressure at the bottom of Well A if the well of 9.1 ppg mud?



Hydrostatic Pressure

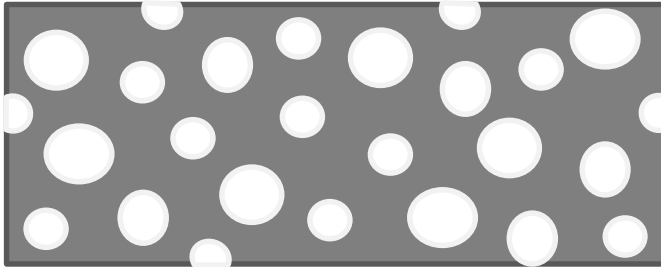
- Quiz

If the fluid level of a 9.9 ppg fluid in the wellbore fell by 700 ft, how much would the hydrostatic pressure be reduced?

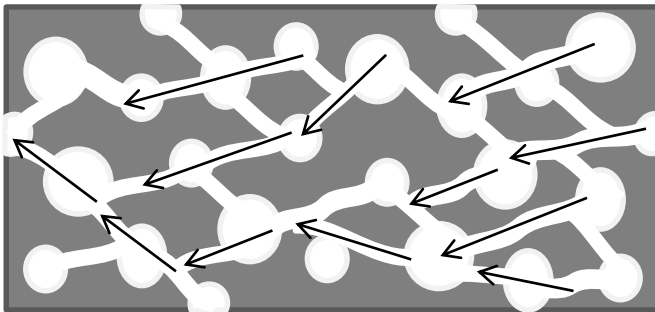


Formation Pressure

- Formation pressure or Pore Pressure is the pressure contained in the pore spaces of a rock.
- Porosity (storage capacity)

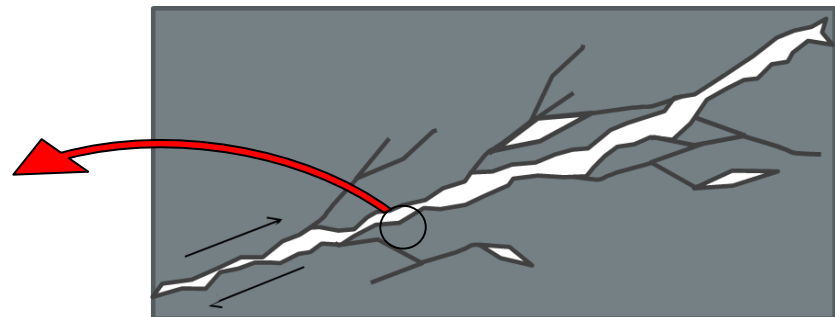


- Permeability (flow capacity)



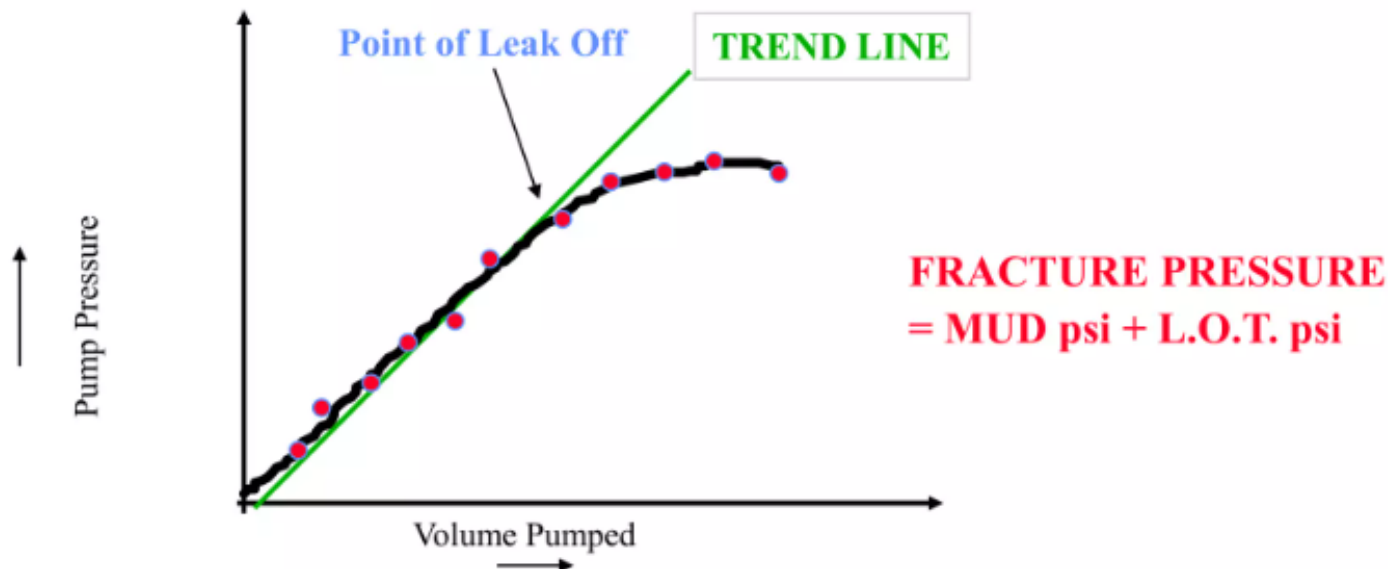
For drilling control [well stability] and for well control the **hydrostatic pressure** must be at least equal but preferably **greater** than the Formation pressure.

Fracture permeability

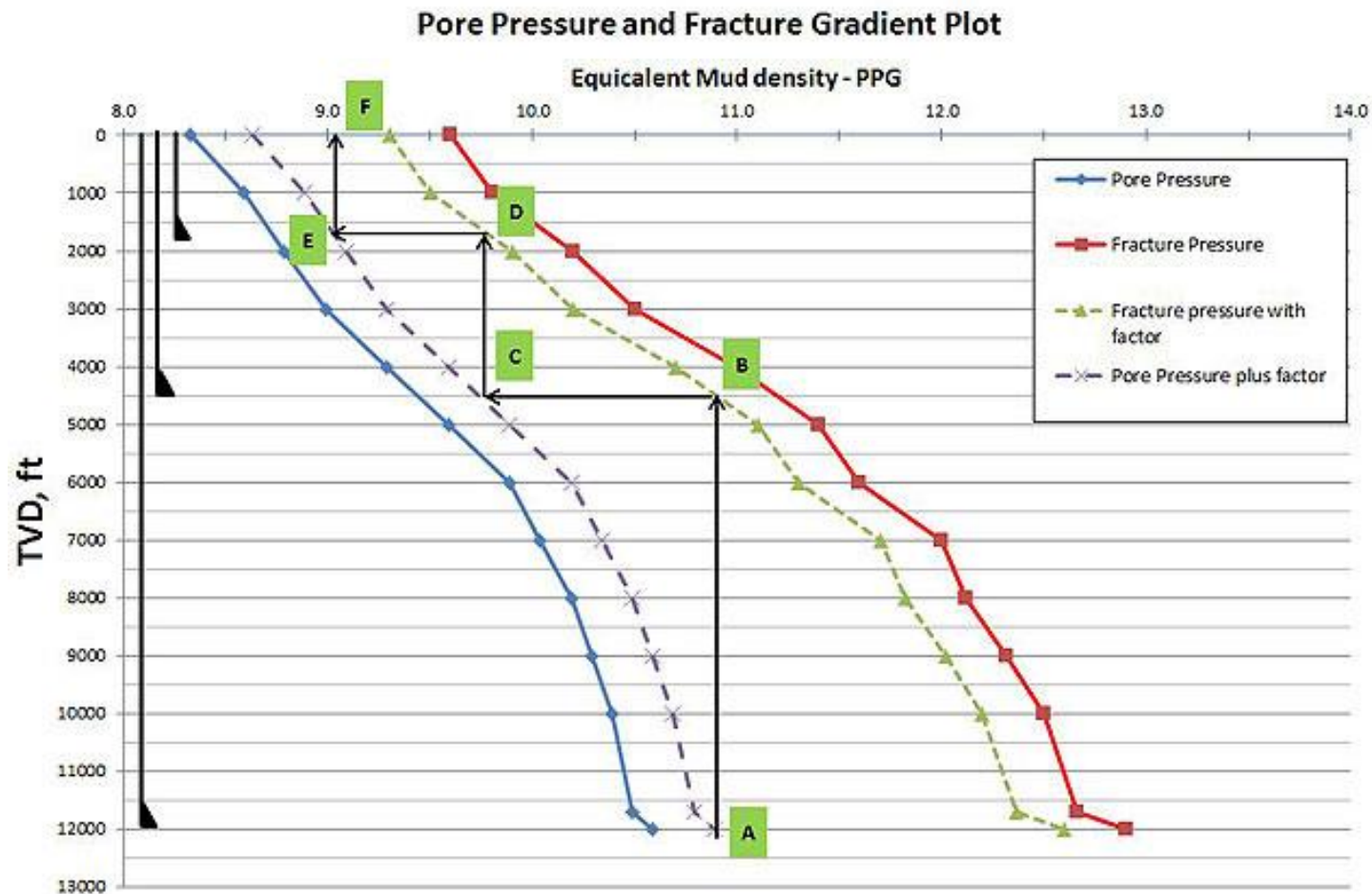


Fracture Pressure

- Fracture pressure is the amount of borehole pressure a formation can withstand before it fails or splits.
- Fracture pressure is normally expressed as a fracture gradient or an equivalent density with units of psi/ft or ppg.
- The test to determine the fracture pressure of the formation is called Leak-off Test (LOT).



PP-FG Plot : Casing Seat



Bottom Up Design

<https://www.drillingformulas.com/casing-seat-selection-how-to-select-casing-setting-depth/>

Formation
Pressure

<

Mud Hydrostatic

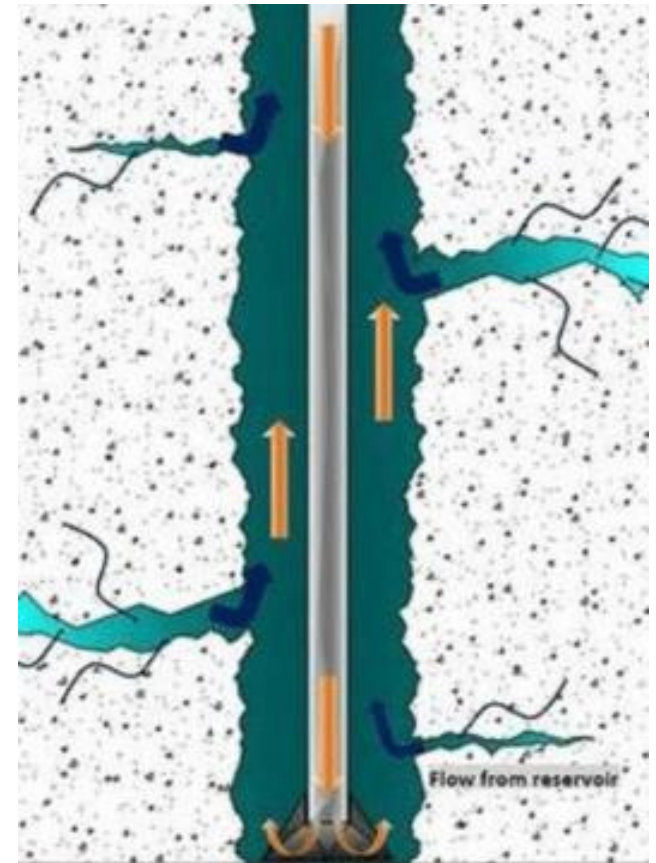
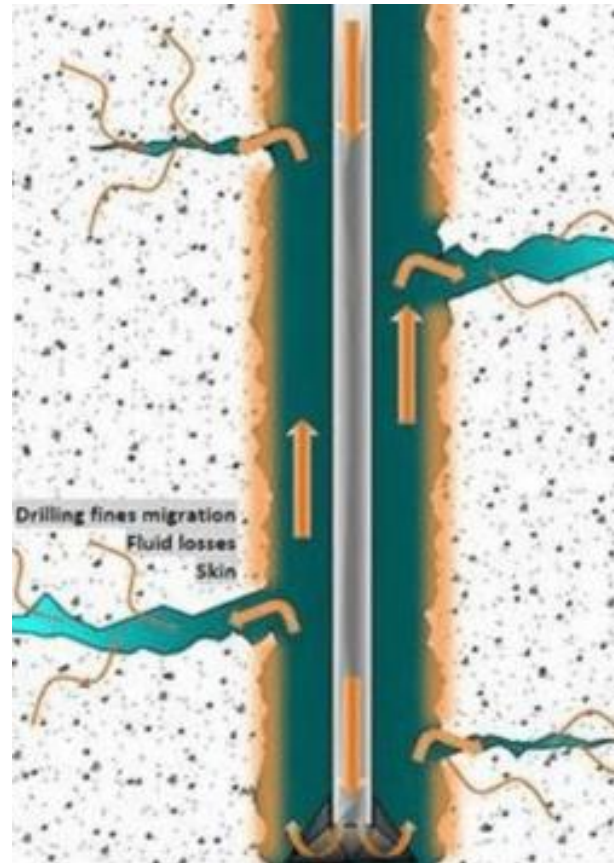
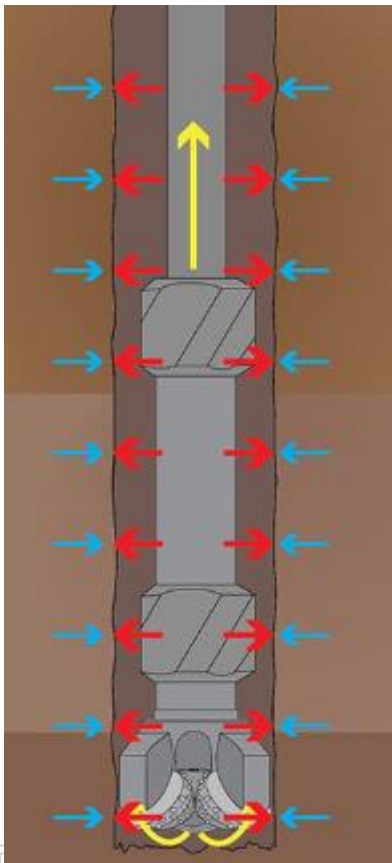
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Fracture
Pressure

± Balance

>> Overbalanced

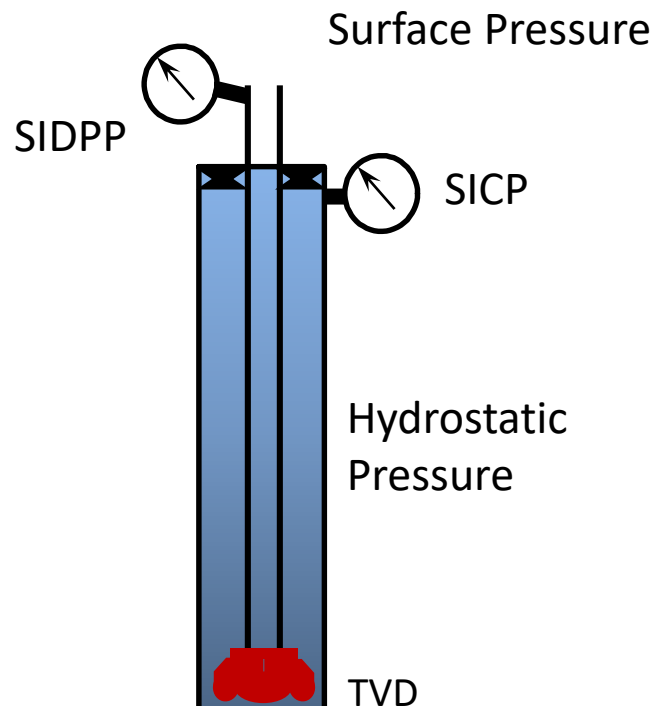
<< Underbalanced



Surface Pressure

- We use the term surface pressure to describe any pressure that is exerted at the top of a column of fluid.

Well shut-in condition – due to kick



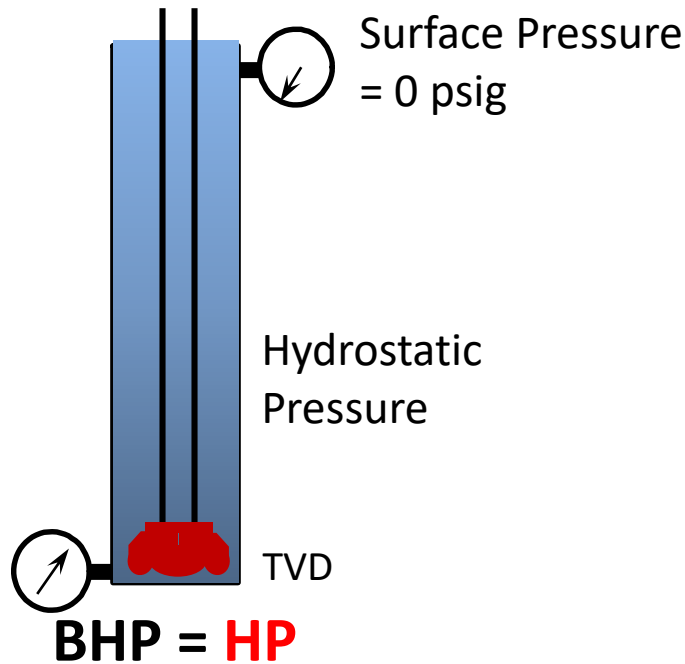
Surface Pressure Terminology:

1. SIDPP = Shut-In Drill Pipe Pressure
2. SICP = Shut-In Casing Pressure

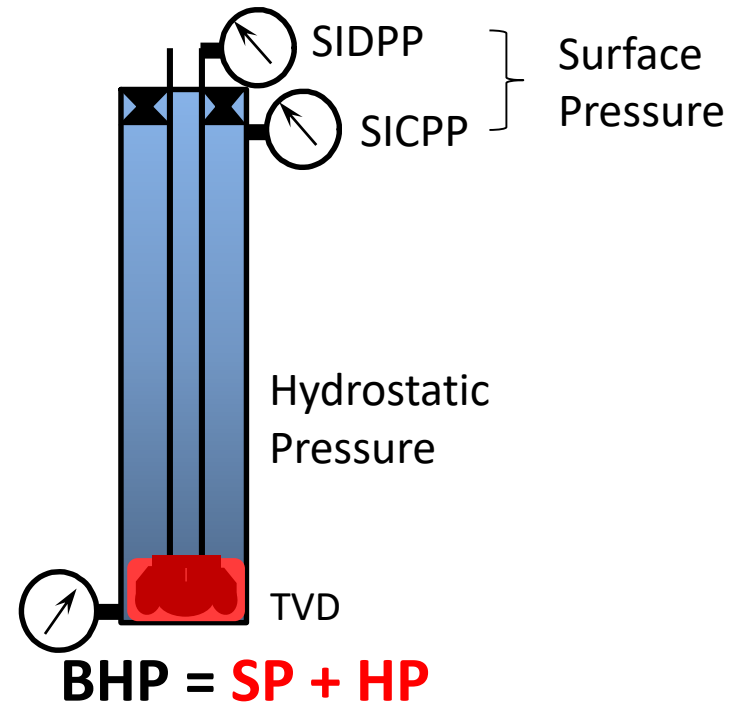
Bottom Hole Pressure

Static Condition: No fluid and pipe movement

Well open to atmospheric condition



Well shut-in condition – due to kick



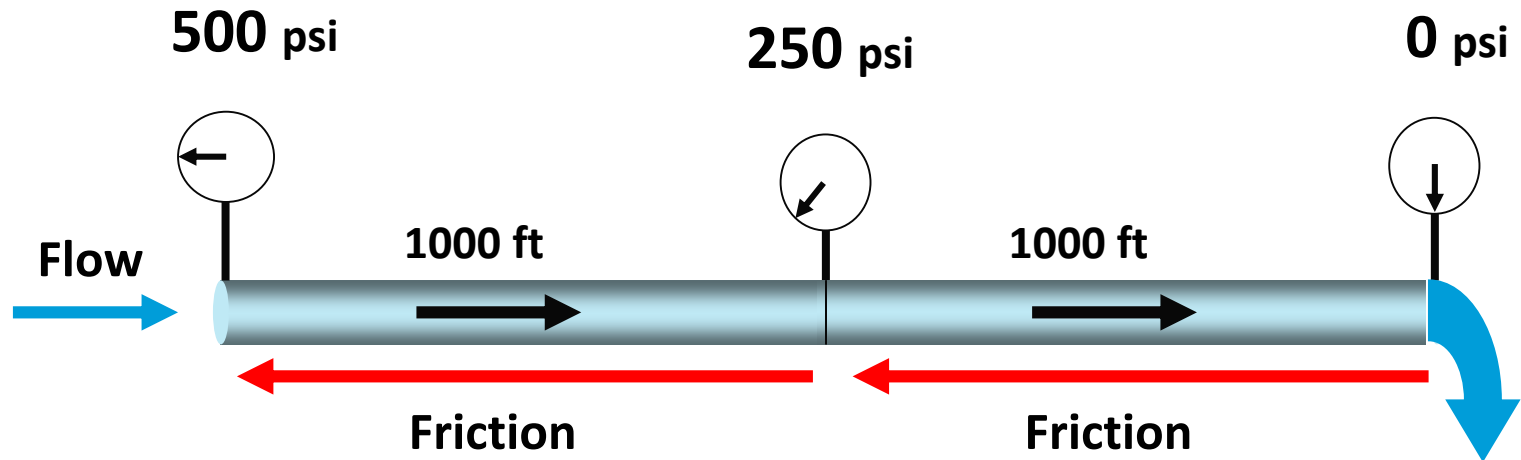
BHP = Formation Pressure, after shutting in well after a kick and with the bit on bottom

In Drill String the BHP = HP mud in String + Shut-In Drill Pipe Press. (SIDPP)

In Casing/Open Hole the BHP = HP mud in annulus + HP Kick + Shut-In Casing Press. (SICP)

Friction Pressures (FrP)

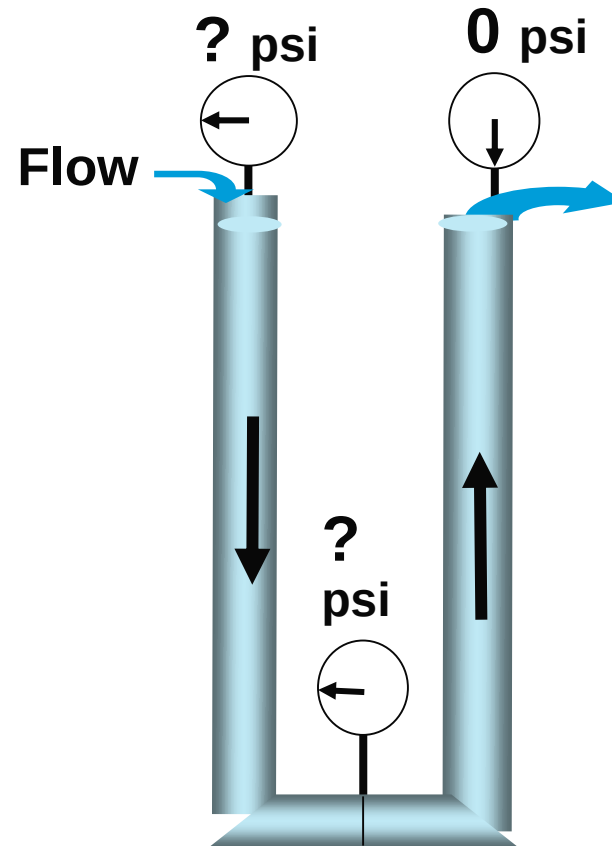
Dynamic condition: when fluid is not static and flowing, there is a *pressure that opposes to the flow*, called **Friction Pressure**.



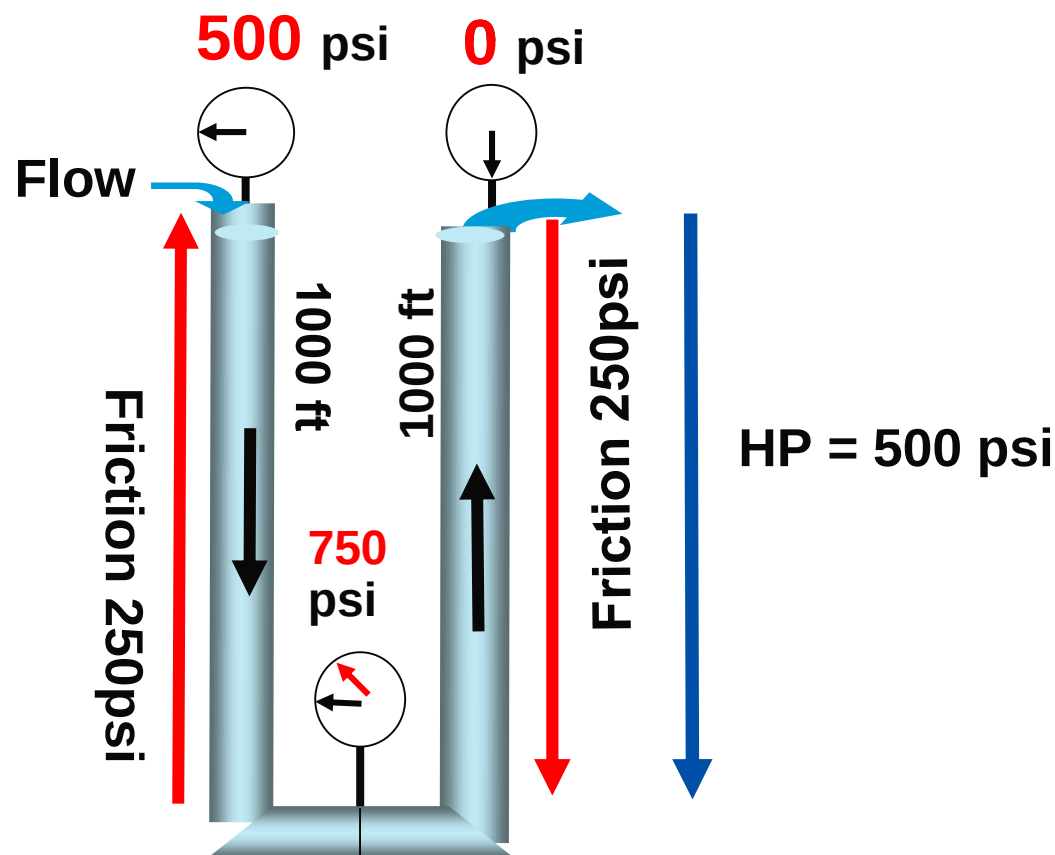
A gauge will see all the friction pressures **downstream**.

Friction Pressures (FrP)

- Friction Acts in Opposite Direction to Flow



Friction Pressures (FrP)



$$\text{DPP} - \text{FrP}_{\text{ds}} + \text{HP}_{\text{ds}} = \text{BHP} = \text{CP} + \text{FrP}_{\text{ann}} + \text{HP}_{\text{ann}}$$

$$500 \text{ psi} - 250 \text{ psi} + 500 \text{ psi} = 750 \text{ psi} = 0 \text{ psi} + 250 \text{ psi} + 500 \text{ psi}$$

Friction Pressure (FrP)

Geometry

- Length (MD)
- Flow area (size and shape)

Fluid Properties

- Rheology
- Density (approximate linear relationship)

$$\text{FrP}_2 = \text{FrP}_1 \times \frac{\text{MW}_2}{\text{MW}_1}$$

Flow Rate (approximate squared relationship)

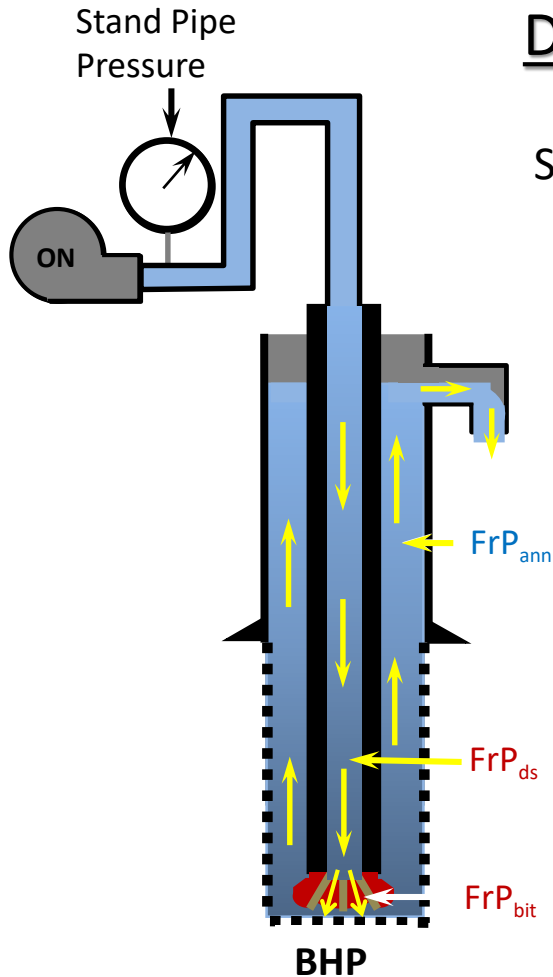
$$\text{FrP}_2 = \text{FrP}_1 \times \frac{(Q_2)^2}{(Q_1)^2}$$

Friction Pressure (FrP)

What is the new friction pressure in the system if the mud weight is increased from 9.5 ppg to 10.5 ppg, and the rest of the parameters remain the same? The current friction pressure in the system is 700 psi.

- A. 700 psi
- B. 774 psi
- C. 633 psi
- D. 733 psi

Bottom Hole Pressure



Dynamic Conditions (Forward Circulation):

$$\text{Standpipe pressure} = FrP_{\text{surf}} + FrP_{\text{ds}} + FrP_{\text{bit}} + FrP_{\text{ann}}$$

- Friction pressure acts in opposite direction to Flow
- FrP_{ann} is the only Friction Pressure that acts downwards.
- Therefore, BHP will increase by FrP_{ann}

$$\text{BHP} = \text{HP} + FrP_{\text{ann}}$$

- BHP expressed in ECD will be:

$$\text{ECD} = \text{MW} + \frac{FrP_{\text{ann}}}{0.052 \times \text{TVD}}$$

ECD is the expression of BHP in equivalent MW.

Bottom Hole Pressure

Calculate the ECD in forward calculation (ignore the surface line friction):

- Fluid weight: 12.0 ppg
- Depth: 10,000 ft MD/8,000 ft TVD
- Pump pressure: 3,500 psi
- Friction pressure in DP: 1,000 psi
- Friction pressure across the bit: 2,100 psi

- A. 12.8 ppg
- B. 13.0 ppg
- C. 13.9 ppg
- D. 19.5 ppg

Relationship of Pressure to Volume

- Gas is **lighter** than a fluid and will **migrate** when well is still shut in, creating high[er] shut-in surface pressures if not expanded!
- Volume of gas must be allowed to **expand** in order to drop its pressure as it comes to surface.
- Gas is **compressible** and will respond to changes in pressure and temperature.

❑ Ideal Gas Law: **$P \times V = n \times R \times T$**

❑ We ignore the effect of temperature for now

- This means:

❑ As VOLUME increases, PRESSURE decreases.

❑ As VOLUME decreases, PRESSURE increases.

Therefore we can say: **$P \times V = C$** [Constant]



processtechacademy.com

Relationship of Pressure to Volume

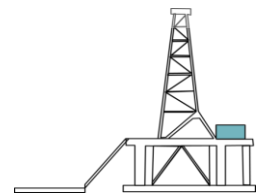
- Example:
 - Gas Kick Volume $[V_1]$ is 4 bbls.
 - Formation gas pressure $[P_1]$ is 4200 psi.
 - Gas migrates but is also allowed to expand by opening the choke.
 - We have bled off 2 bbls from the choke; the new gas volume is therefore $4 + 2 = 6$ bbls $[V_2]$

- Question:

What will be the new gas pressure after expansion $[P_2]$?

- Answer:

Kick Causes, Detection, and Shut-In



- **What is “KICK”?**
 - **What are the CAUSES of KICKS?**

Kick Definition

A kick is an undesirable and unintentional flow of formation fluid into the well bore.

Causes of Kick:

- Abnormal pressure
- Failure to fill the hole properly
- Insufficient Fluid Density
- Lost circulation
- Swabbing
- Human error
- Annular gas flow after cementing

Possible Kick Indicators

Real time:

- ROP increase or decrease (drilling break)
- Change in drilling parameters (torque and drag)
- Decrease in Shale Density

Bottoms-up:

- Increase in connection/trip/background gas (possible loss of ECD)
- Changes in cuttings volume/size/shape
- Changes in mud properties (gas cut mud, water cut mud)
- Changes in flowline temperature

Kick Detection -- Alarms:

- High and low pit level alarms — audio and visual (e.g. set at +/- 5 bbl *)
- Flow returns indicator — audio and visual

Positive Kick Indicators

- While drilling:

Prescribed Action	→	
○ Increase in flow		Driller must shut in
○ Pit gain		
○ Flow with the pumps off		

- While tripping:

Prescribed Action	→	
○ <u>Positive flow</u> with pipe static		Driller must shut in
Prescribed Action	→	
○ Trip log deviation:		Driller must check for flow :
• Short fill ups tripping out • Excess pit gain tripping in		• If <u>flowing</u>, shut in well. • If not flowing, prepare to trip to bottom.

Shut-in Procedures

- While drilling (three S's):
 - **S**pace out (with respect to BOPE)
 - **S**hut down (stop rotation and stop pumping)
 - **S**hut-in the well
- While tripping (three S's):
 - **S**tab drill pipe safety valve (close same)
 - **S**pace out (with respect to BOPE)
 - **S**hut-in

Note: Install IBOP and prepare to strip back to the bottom.



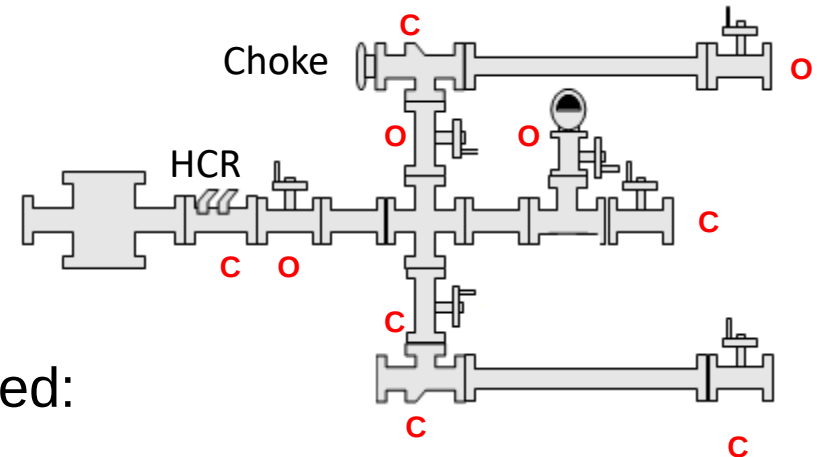
Shut-in Types & Sequence - API

Hard shut-in: Choke & HCR closed:

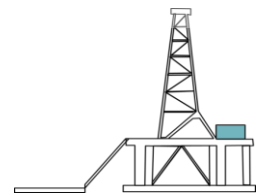
- Close preventer
- Open HCR

Soft shut-in: Choke open & HCR closed:

- Open HCR
- Close preventer
- Close choke



Constant BOTTOM HOLE PRESSURE Concepts



- What is Constant BHP concept?
 - What are the Well Control Methods that use the Constant BHP concepts?

Constant BHP Concepts

- Correctly identify the surface pressures that balance the formation pore pressure.
- The constant BHP techniques of:
 - Obtaining shut-in casing pressure (SICP)
 - Obtaining shut-in drill pipe pressure (SIDPP) and Kill Mud Weight (KMW)
 - Selecting a kill rate and bringing pumps up to speed
 - Obtaining initial circulating pressure (ICP) and slow circulating rate (SCR)
 - Using the pump and choke to maintain constant BHP
 - Accounting for lag time when manipulating the choke

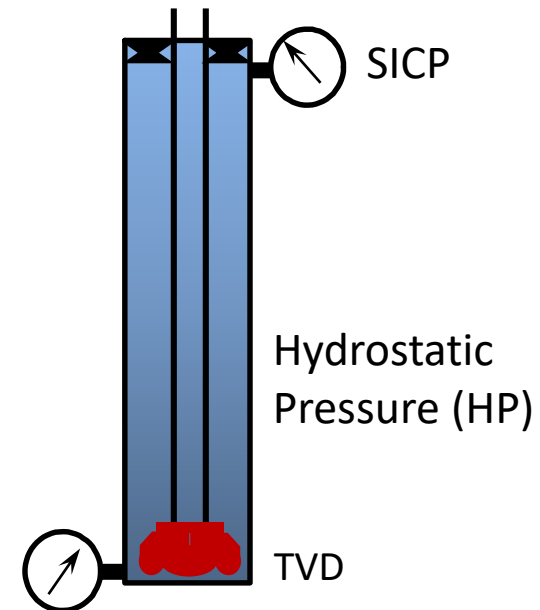
Obtaining Shut-in Casing Pressure (SICP)

- SICP is a measure of the difference between the formation pressure and the HP in the annulus when a kick occurs:

$$\text{SICP} = \text{FP} - \text{HP}_{\text{ann mud}} - \text{HP}_{\text{influx}}$$

- Procedure for obtaining SICP:
 1. Confirm shut-in well.
 2. Open HCR valve and begin recording casing pressures.
 3. Record SICP when CP has **stabilized**.

Well shut-in condition – due to kick



BHP = Formation Pressure, after shutting in well after a kick and with the bit on bottom

Shut-in Drill Pipe Pressure (SIDPP)

- SIDPP is a measured difference between the BHP and the HP of the fluid column in the drillstring:
 - Obtained once casing pressure has stabilized
 - Must know mud weight inside of DP
- SIDPP gives a direct reading of the formation pressure during a kick:

$$\text{SIDPP} = \text{FP} - \text{HP}_{\text{drill string}}$$

- Kill Mud Weight (KMW) can then be calculated using SIDPP:

$$\text{KMW(ppg)} = \text{Original MW(ppg)} + \frac{\text{SIDPP(psi)}}{0.052 \times \text{TVD (feet)}}$$

Starting the Operation —Surface Stack

Bring Pumps Up To Kill Speed holding **CP** constant:

- Always hold CP constant when starting/stopping/changing pump speeds to keep BHP constant.
- Make sure CP does not drop below SICP.
- Once CP has stabilized, allow for lag time before observing circulating DPP.

Static ➡ $\overline{\text{BHP}} = \overline{\text{HP}}_{\text{Ann}} + \overline{\text{CP}}$

Dynamic ➡ $\overline{\text{BHP}} = \overline{\text{HP}}_{\text{Ann}} + \overline{\text{CP}} + \cancel{\overline{\text{FrP}}_{\text{Ann}}}$

Hold constant with choke
BHP relatively **constant**.



Initial Circulation Pressure (ICP)

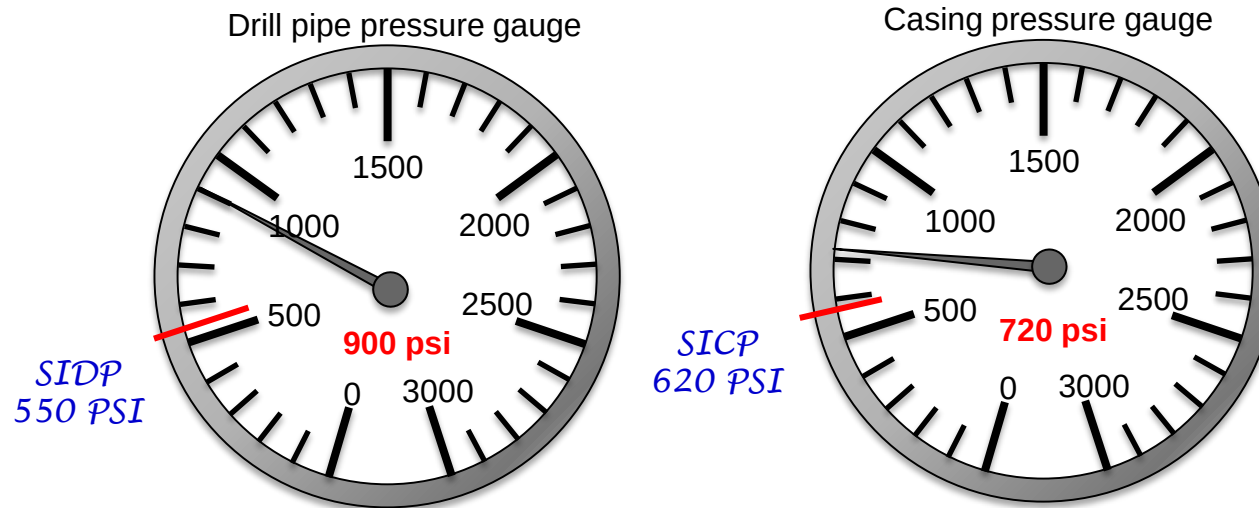
- The circulating pressure resulting after pump reach kill speed is called initial circulating pressure (ICP).
- ICP is the sum of SIDPP and pressure required to overcome friction:

$$\text{ICP} = \text{SIDPP} + \text{SCR}$$

- SCR pressure is the pressure required to overcome **friction** at a slow pump rate.
- By definition, ICP does NOT include a **safety factor / overbalance**.
- When circulating, if DP pressure goes below ICP, then the well will be **underbalanced** and an additional influx is possible.

Initial Circulation Pressure (ICP)

What is the ICP after pump reach kill speed?
Pre-recorded SCR Press = 250 psi



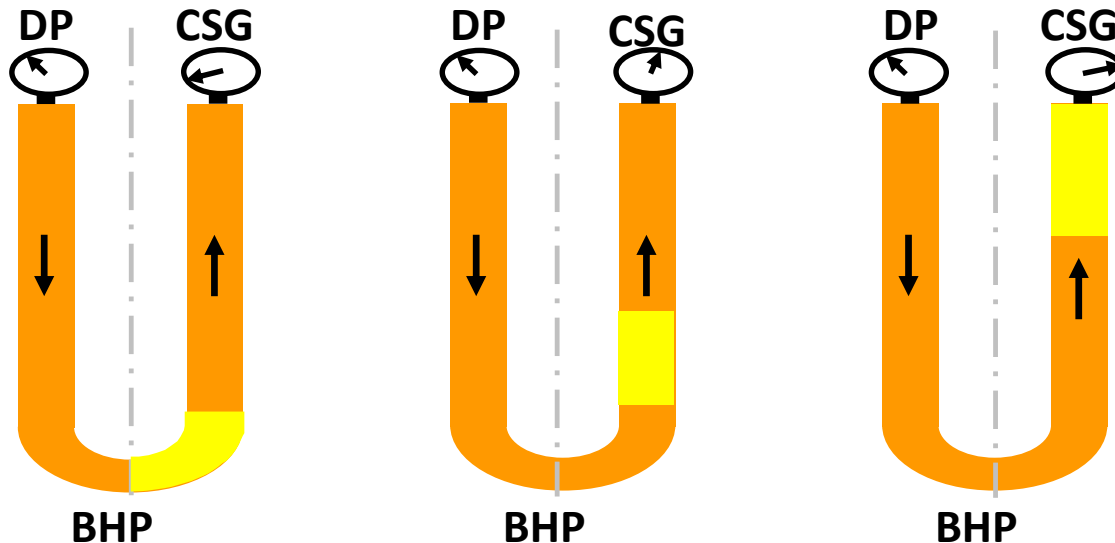
- A. 720 psi
- B. 800 psi
- C. 870 psi
- D. 970 psi

Driller's Method

- Constant BHP well control method
- Using two (2) circulations:
 - First circulation **removes influx** with the original fluid.
 - Second circulation **kills the well** with kill fluid.

Driller's Method First Circulation

Removing the influx



$$\overline{\text{DPP}}_{\text{DS}} + \overline{\text{HP}}_{\text{DS}} - \overline{\text{FrP}}_{\text{DS}} = \overline{\text{BHP}} = \downarrow \text{HP}_{\text{ann}} + \uparrow \text{CP}_{\text{ann}} + \cancel{\text{FrP}_{\text{ann}}}$$



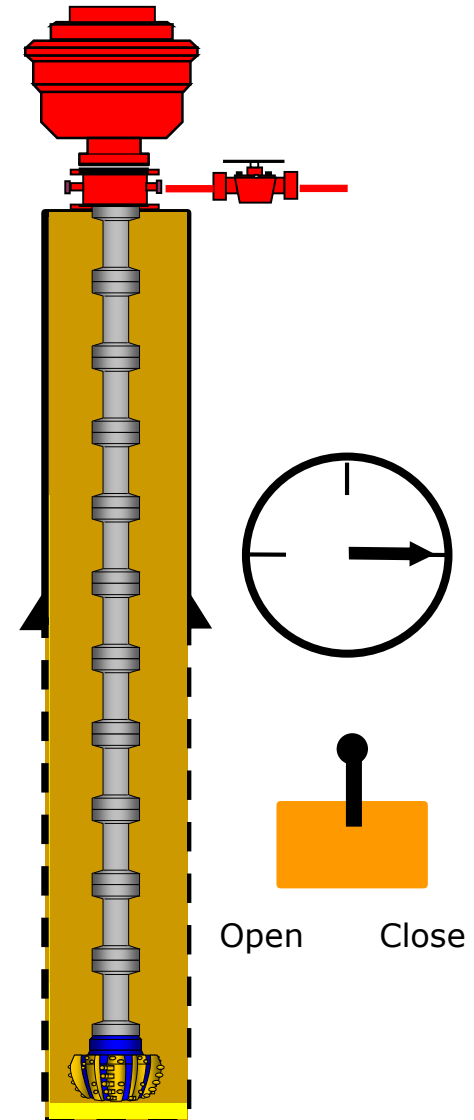
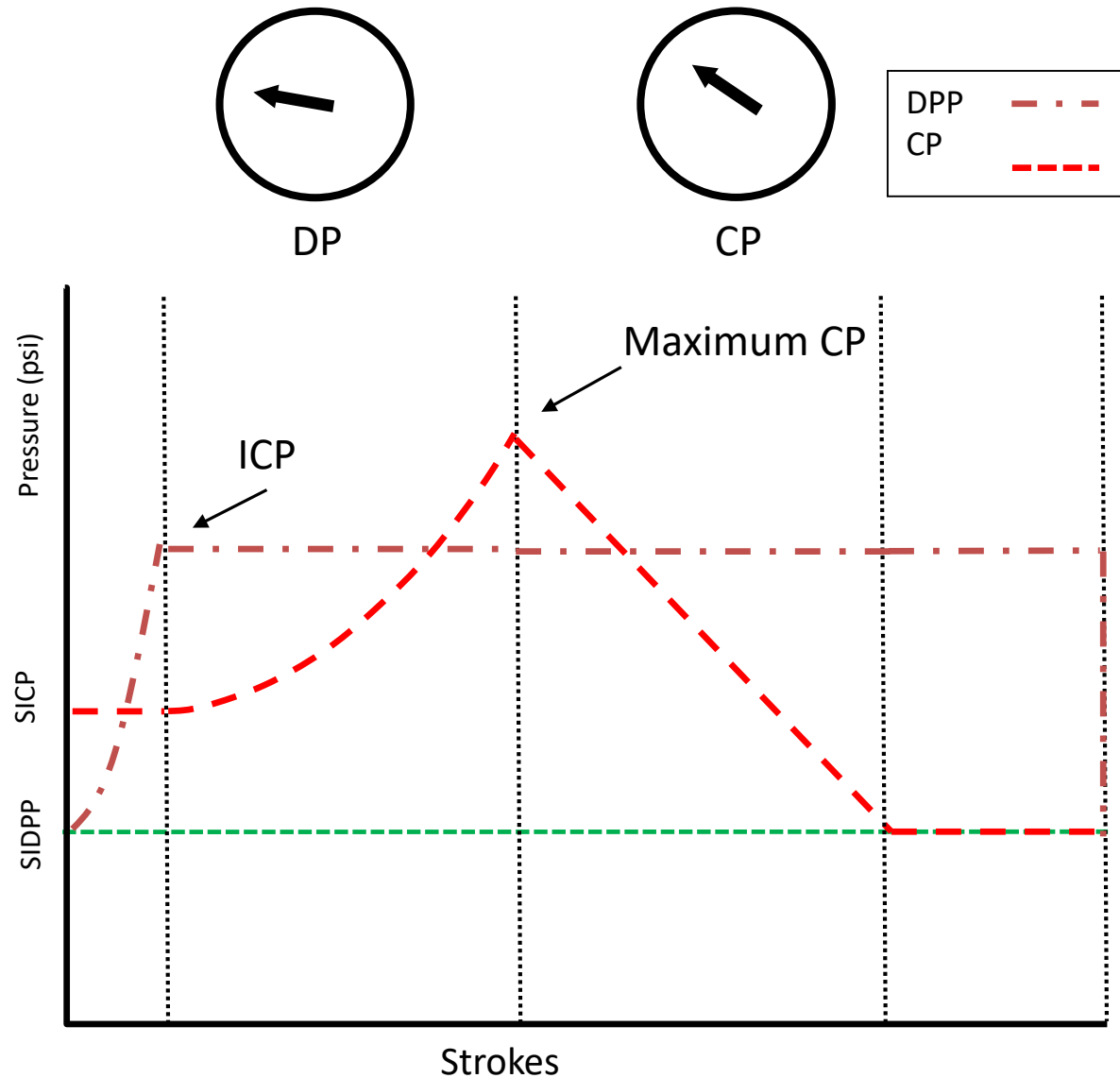
Hold **DPP** constant with the choke.

Driller's Method

First Circulation Procedure

1. Start up pump and bring to kill speed holding **CP** constant.
2. Read and record ICP at DP pressure gauge
3. Hold DPP constant until influx is circulated out.
4. Shut down pumps holding **CP** constant.
5. Compare SIDPP and SICP.

Driller's Method First Circulation Chart



Driller's Method Second Circulation

The purpose of the second circulation is to kill the well.

Calculate and mix KMW:

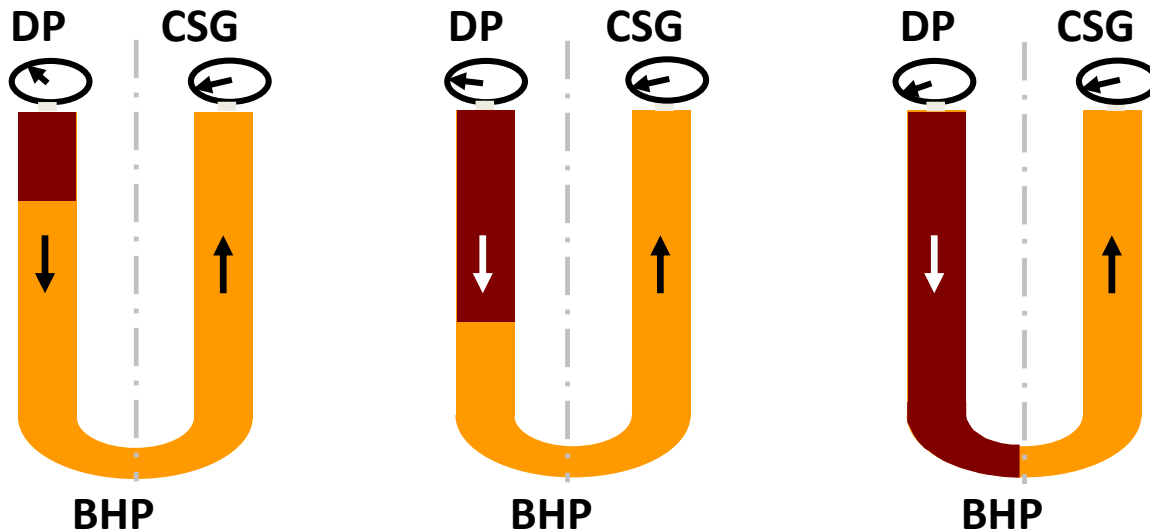
- $$\text{KMW (ppg)} = \text{Current MW (ppg)} + \frac{\text{SIDPP (psi)}}{0.052 \times \text{TVD (feet)}}$$

Calculate strokes from surface to bit:

- $$\text{Drill String Cap (stks)} = \frac{\text{DP}_{\text{cap}}(\text{bbl/ft}) \times \text{MD}(\text{feet})}{\text{Pump output}(\text{bbl/stks})}$$

Driller's Method Second Circulation, (Continued)

Displace KMW.



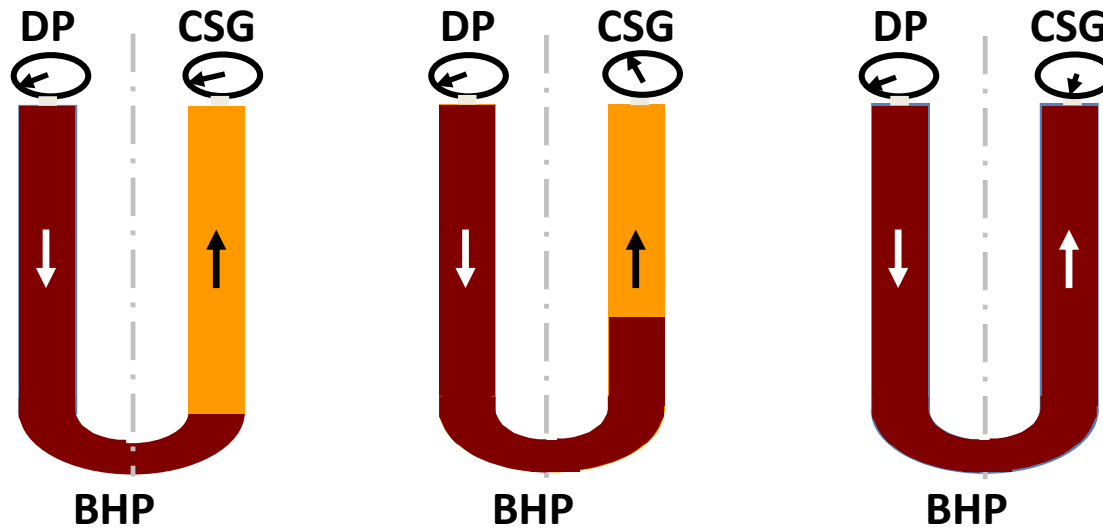
$$\downarrow \text{DPP}_{\text{DS}} + \uparrow \text{HP}_{\text{DS}} - \uparrow \text{FrP}_{\text{DS}} = \text{BHP} = \text{HP}_{\text{ann}} + \text{CP}_{\text{ann}} + \cancel{\text{FrP}_{\text{ann}}}$$

Hold CP constant while pumping 
KMW to bit

This method is simple, but requires no gas left in the annulus.

Driller's Method Second Circulation, (*Continued*)

While circulating KMW from bit to surface:



$$\overline{\text{DPP}}_{\text{DS}} + \overline{\text{HP}}_{\text{DS}} - \overline{\text{FrP}}_{\text{DS}} = \overline{\text{BHP}} = \uparrow \text{HP}_{\text{ann}} + \downarrow \text{CP}_{\text{ann}} + \cancel{\text{FrP}_{\text{ann}}}$$

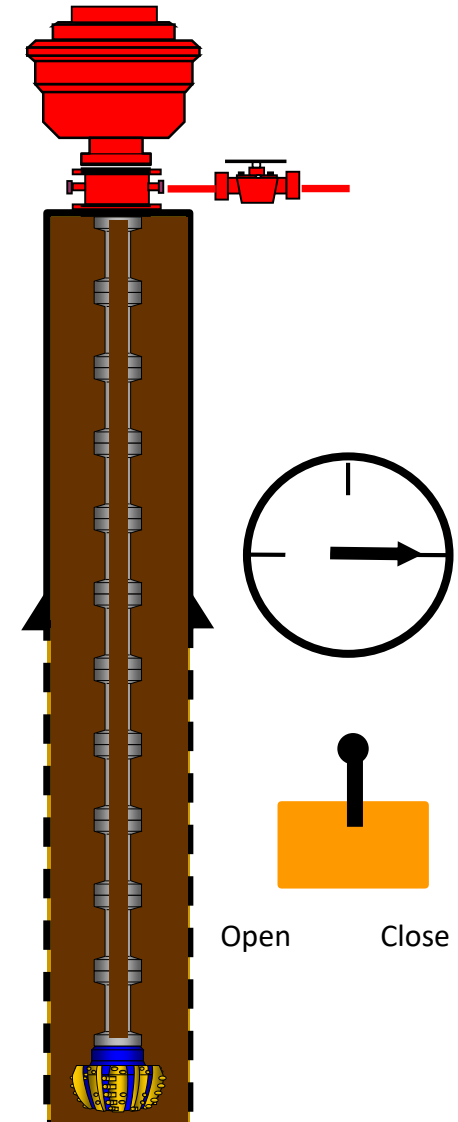
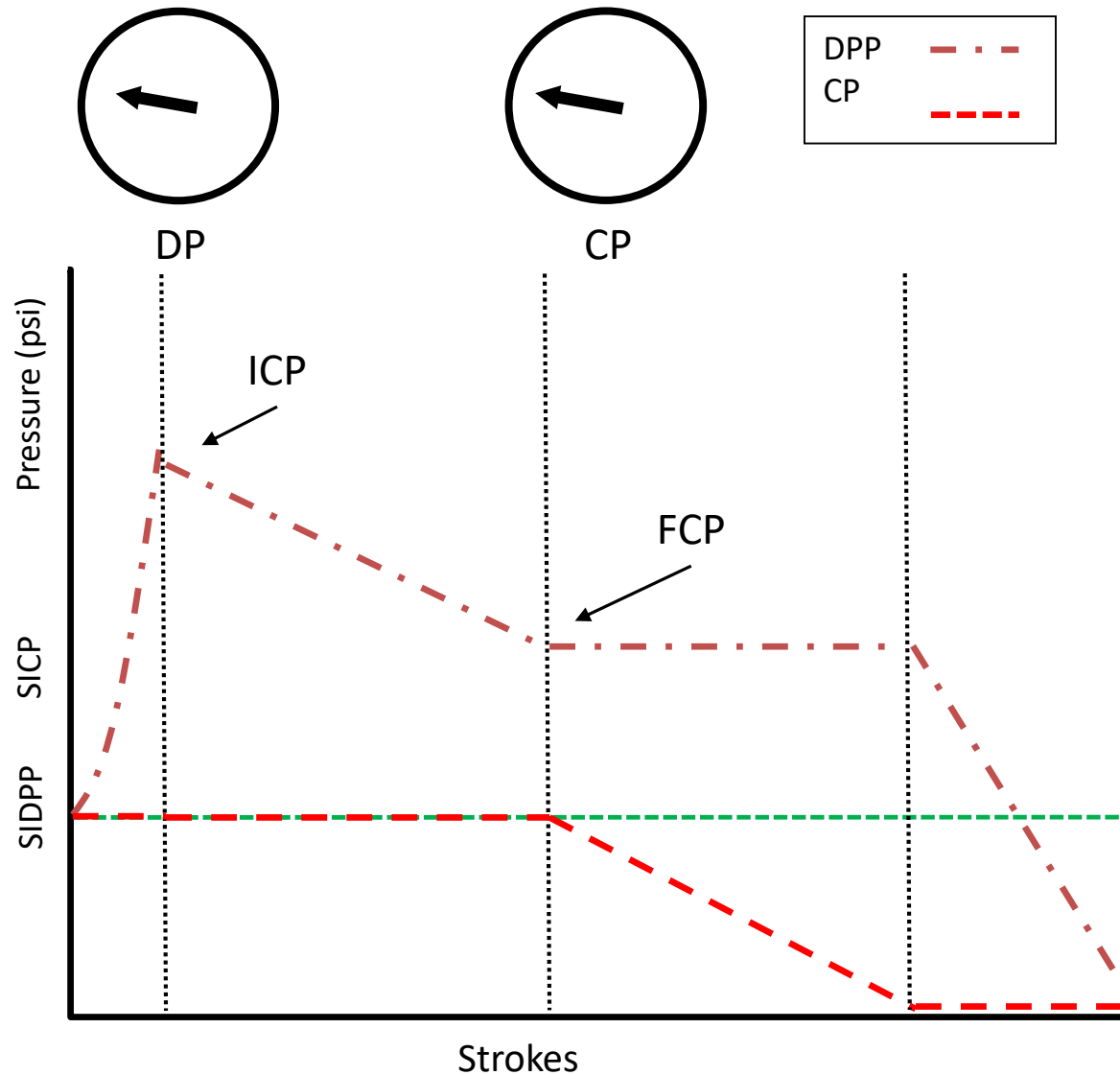
↑ Hold DPP constant at **final circulation pressure**.

Driller's Method

Second Circulation Procedure

1. Start up pump and bring to kill speed holding **CP** constant.
2. Hold CP constant until KMW reaches the bit.
3. Read and record DPP (FCP), once KMW reaches the bit.
4. Hold DPP constant at FCP until the total system is displaced and $MW_{in} = MW_{out} = KMW$.
5. Shut down pumps holding CP constant.
6. Shut-in and confirm well is dead.

Driller's Method Second Circulation





Thank You

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